

Mar 11, 2026

Statement of Work (SOW)

Joint Base McGuire-Dix-Lakehurst (JB MDL) NJ

SONET Replacement Project.

1. INTRODUCTION. This SOW defines the requirement for the Contractor to Engineer, Furnish, Install and Test (EFI&T) into service, a replacement of the current Cisco Systems 15454 SONET Ring. **This Statement of Work covers procurement and installation activities only. No ongoing support, maintenance, or operational services are included or required as part of this SOW.** The current SONET Ring does not ride the base network – it utilizes the base fiber plant only for interconnectivity. See Figure 1 and Table 1 below. There are Eight (8) locations spread across the JB MDL. Buildings (B) 1901 RDP Room, 1730 ITN Room, and 2217 Switch Room on McGuire proper. B 3369 Switch Room, B 5321 Dial Central Office (DCO), B 5632 ITN Room, and B 5656 Room 1311 on Ft. Dix proper. And Hangar 1 Switch Room on Lakehurst NAS. This project shall replace the existing manufacturer discontinued Cisco 15454 SONET equipment with an Optical Transport Network (OTN) equipped with (e/w) modules required to meet the Table 1 circuits. For the rest of this document, the OTN shall be referred to as the SONET Replacement System (SRS). This SRS shall not be on the base network but, like the current SONET Ring, only utilize base fiber (Layer 1) for interconnectivity. All future references to Government Furnished shall be reflected as (GFE). All locations are on the JB MDL, NJ. The Government will designate a Point of Contact (POC) and Project Point of Contact (PPOC) for this effort. The assigned POC for this contract is Mr. Brian Uldschmidt. The assigned PPOC for this contract is Kaitlin Diaz.

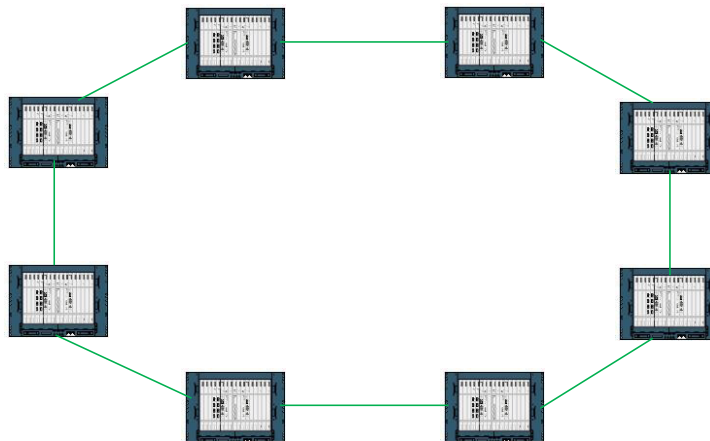


Figure 1: Current JB MDL SONET Ring

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15454 Circuit Designator	Source	Destination	Circuit Description
VT1.5	B1901	B 2217	T-1 Time Domain Multiplex (TDM) 1.544 Mb/s
STS-12c	B1901	B 5321	OC-12 (644Mb/s) with Cable Connection is SM Fiber
VT1.5	B1901	B 5656	T-1 TDM 1.544 Mb/s Cable connection is CAT 5e
STS-1	B1901	B 5632	50 Mb/s Cable Connection is SM Fiber
STS-12c	B1901	B 2217	OC-12 (644Mb/s) with Cable Connection is SM Fiber
STS-3c	B1901	B 5321	OC-3 (155Mb/s) with Cable Connection is SM Fiber
VT1.5	B1901	HANGAR-1	T-1 TDM 1.544 Mb/s Cable connection is CAT 5e
STS-1	B1901	B 5321	50 Mb/s Cable Connection is SM Fiber
STS-1	B1901	HANGAR-1	50 Mb/s Cable Connection is SM Fiber
STS-1	B1901	B 2217	50 Mb/s Cable Connection is SM Fiber
VT1.5	B1901	B 2217	T-1 TDM 1.544 Mb/s Cable connection is CAT 5e
STS-1	B1901	B 2217	50 Mb/s Cable Connection is SM Fiber
VT1.5	B1901	B 2217	T-1 TDM 1.544 Mb/s Cable connection is CAT 5e
VT1.5	B1901	B 2217	T-1 TDM 1.544 Mb/s Cable connection is CAT 5e
VT1.5	B1901	B 2217	T-1 TDM 1.544 Mb/s Cable connection is CAT 5e
VT1.5	B1901	HANGAR-1	T-1 TDM 1.544 Mb/s Cable connection is CAT 5e
VT1.5	B1901	B 5321	T-1 TDM 1.544 Mb/s Cable connection is CAT 5e
VT1.5	B1901	B 5321	T-1 TDM 1.544 Mb/s Cable connection is CAT 5e
VT1.5	B1901	B 5321	T-1 TDM 1.544 Mb/s Cable connection is CAT 5e
VT1.5	B1901	B 5321	T-1 TDM 1.544 Mb/s Cable connection is CAT 5e
STS-3c	B1901	HANGAR-1	OC-3 (155Mb/s) with Cable Connection is SM Fiber
VT1.5	B1901	HANGAR-1	T-1 TDM 1.544 Mb/s Cable connection is CAT 5e
VT1.5	B1901	HANGAR-1	T-1 TDM 1.544 Mb/s Cable connection is CAT 5e
STS-3c	B1901	B 5656	OC-3 (155Mb/s) with Cable Connection is SM Fiber
STS-3c	B1901	B 5656	OC-3 (155Mb/s) with Cable Connection is SM Fiber
STS-12c	B1901	B 5632	OC-12 (644Mb/s) with Cable Connection is SM Fiber
STS-12c	B1901	B 5632	OC-12 (644Mb/s) with Cable Connection is SM Fiber
VT1.5	B1901	B 2217	T-1 TDM 1.544 Mb/s Cable connection is CAT 5e
STS-1	B1901	B 1730	50 Mb/s Cable Connection is SM Fiber
STS-1	B1901	B 2217	50 Mb/s Cable Connection is SM Fiber
STS-1	B1901	B 5321	50 Mb/s Cable Connection is SM Fiber
STS-1	B1901	B 5321	50 Mb/s Cable Connection is SM Fiber
STS-1	B1901	B 5632	50 Mb/s Cable Connection is SM Fiber
STS-1	B1901	B 5632	50 Mb/s Cable Connection is SM Fiber
STS-1	B1901	HANGAR-1	50 Mb/s Cable Connection is SM Fiber

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STS-1	B1901	HANGAR-1	50 Mb/s Cable Connection is SM Fiber
STS-1	B1901	HANGAR-1	50 Mb/s Cable Connection is SM Fiber
STS-1	B1901	B 5632	50 Mb/s Cable Connection is SM Fiber
STS-1	B1901	B 5632	50 Mb/s Cable Connection is SM Fiber
VT1.5	B1901	HANGAR-1	T-1 TDM 1.544 Mb/s Cable connection is CAT 5e
VT1.5	B1901	B 3369	T-1 TDM 1.544 Mb/s Cable connection is CAT 5e
STS-1	B1901	B 5321	50 Mb/s Cable Connection is SM Fiber
15454 Circuit Designator	Source	Destination	
STS-1	B1901	HANGAR-1	50 Mb/s Cable Connection is SM Fiber
VT1.5	B1901	B 3369	T-1 TDM 1.544 Mb/s Cable connection is CAT 5e
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VT1.5	B1901	B 3369	T-1 TDM 1.544 Mb/s Cable connection is CAT 5e
STS-3c	B1901	B 5321	OC-3 (155Mb/s) with Cable Connection is SM Fiber
			See paragraph 1.1.6. below for Analog Circuit requirements

Table 1. Cisco 15454 SONET Circuit List

1.1. SCOPE.

1.1.1. The Contractor shall EFI&T an SRS serving the Eight (8) each locations. The Contractor shall EFI&T their SRS to the appropriate GFE CAT 5e, or Thin Net COAX, or Fiber Optic Patch Panels (FOPP) at each location with the applicable Contractor EFI&T patch cords. All GFE FOPP terminations are Snap-and-Twist (ST) connectors. At all locations, GFE cabinet (Except for B 5632, see below paragraph 3.4.) rack space is available for the Contractor to EFI&T mount their equipment. Each GFE cabinet has spare 110VAC, 60Hz recessed female outlets for the Contractor to EFI&T their SRS's power connections or the required Contractor EFI&T UPS. The Contractor shall EFI&T patch cables from these cabinets to the official GFE Demarcation Point, within the serving Comm Room(s). The Government shall provide cut sheets 10 days after contractor award, detailing circuit terminations for each location and the respective patch panels. **The Government requires a commercial, carrier-grade SONET/optical transport solution compatible with existing JB MDL network infrastructure. A Cisco-based system is highly preferred due to established interoperability, life cycle support, and existing configuration baselines; however, equivalent solutions meeting all technical and performance requirements will be considered. All equipment, components, and materials provided under this contract shall be fully compliant with Trade Agreements Act (TAA) requirements and sourced from TAA-designated countries.**

1.1.2. Note: that if the number of circuits terminating in B 1901 requires an additional SRS expansion shelf, the Contractor shall EFI&T their equipment to meet B 1901 requirements.

If an additional SRS Core Unit is required, the two systems shall be contractor EFI&T to act as one unit. Either scenario shall be reflected in their proposal, their List of Materials, and all required documentation.

1.1.3. Existing GFE fiber optic outside plant connectivity shall be utilized by the Contractor to connect to all buildings in a fully redundant rotating ring. The Government is responsible for all fiber patches linking the 8 locations together within the JB MDL Core Nodes only. At all Eight 3 locations, the Contractor shall EFI&T the required fiber patch cords from their SRS equipment to the GFE FOPP within the serving Comm Room(s). The Contractor shall also EFI&T all required cables for each location's drop circuits (See Table 1). The Government shall provide each location's

fiber cross connections' and drop circuits', cut sheets. The Government shall participate in all cross connections with the Contractor.

1.1.4. It is the Contractor's responsibility to test all their equipment and cables to ensure that they meet all standards and requirements. The Contractor shall correct any deficiency and provide the Point of Contact (POC) and PPOC all testing results. The Contractor shall provide an end-to-end test on all circuits to ensure that there is minimal interference and correct functionality.

1.1.5. Complete building floor plans shall be provided by the Government as part of the Contract Solicitation. These are provided for the Contractor to properly plan their equipment locations and required cabling lengths. Before installing equipment in any cabinet, the Contractor is responsible to coordinate with Government representatives that the cabinet is properly grounded. If it is not, the Government shall take all required actions to ground the cabinet properly.

1.1.6. Per the last Table 1 entry, there are 9 analog circuits from B 1901 to Hangar 1. These are 2 each 2Wire E&M circuits, and 7 each Voice grade circuits which are Plain Old Telephone System (POTS) lines. For the Voice Grade circuits, Hangar 1 shall be Foreign Exchange Office (FXO) interfaces and the B 1901 end shall be Foreign Exchange Subscriber (FXS) interfaces. This requires the Contractor to EFI&T a Circuit to Packet (CTP) Layer II switch at both locations and connect them to the respective Contractor EFI&T SRS with an IP connection. The Government shall provide connection cut sheets and participate in all cross connections with the Contractor.

1.1.7. The Contractor shall EFI&T all materials, and perform all work to include making all cross-connection assignments and changes required to complete requirements as specified in this SOW. The Contractor shall provide a List of Materials to include all equipment to be installed. The Contractor shall coordinate all installations through 87 CS and the Point of Contact. If any electrical work or coordination is required, the Contractor shall notify the PPOC and POC within five (5) business days of identifying the need. Following this notification, the Contractor is responsible for contacting the CE Electrical/Power Pro Work center at 87ces.ceoer.workforce@us.af.mil to request the required support. The Contractor is advised that CE work orders require a minimum of 4–6 weeks to process; therefore, it is critical that all electrical requirements are identified during site surveys to avoid delays.

1.1.8. All Contractor EFI&T SRS equipment shall be on the Defense Information Systems Agency's (DISA) Approved Products List (APL).

2.0. PROJECT SUPPORT REQUIREMENTS.

- 2.1. Project Requirement:** For the areas identified above, the Contractor shall provide all management, personnel, equipment, tools, transportation, supplies, materials, supervision, and other incidentals necessary to EFI&T this replacement. All materials provided shall be new and not refurbished.
- 2.2. Base Support Requirements.** The Contractor shall identify any base support requirements (for example, laydown and storage areas) necessary to complete this project.

2.2.1. The Contracting Officer will assign an 87 CS individual to act in their stead during the installation and designate them as the PPOC. The Contractor shall coordinate all efforts with the PPOC. The PPOC shall process all Contractor efforts and requests with the Contracting Officer and appropriate base personnel.

2.2.2. The Government shall provide 100' square feet of secure and dry storage space for Contractor materials at McGuire AFB – the PPOC shall identify the location. The Contractor shall be responsible for loading, unloading, unpacking and transportation of all Contractor provided materials, test equipment, training documentation, etc.. The Contractor shall be responsible for disposal of all packing materials and unused installation materials. If the PPOC concurs, the Contractor may turn these items over to the Government for use/disposal at no compensation to the Contractor. If the PPOC does not agree, the Contractor shall properly dispose of these items IAW the local environmental practices.

2.2.3. Shipping Address for project materials is: Project POC is:

C/O 87 CS/SCXP
2320 Radin Avenue
McGuire AFB, NJ 08641

Brian Uldschmidt
(609)754-3945
2320 Radin Avenue
McGuire AFB, NJ 08641

Project PPOC is:
Kaitlin Diaz
(609)754-8011
1901 E. Third Street
McGuire AFB, NJ 08641

The Contractor shall send an email to Brian Uldschmidt at brian.uldscmidt.1@us.af.mil and Kaitlin Diaz kaitlin.diaz.1@us.af.mil 5 days prior to shipment. The Contractor shall include tracking information within the email.

2.2.4. Stated work and associated products shall be performed at the UNCLASSIFIED level. However, all work will take place in secure areas where Contractor employees must be always escorted. The Contractor must coordinate access to secure areas at least three (3) working days ahead of time. It is the Government's responsibility to provide escorts. Contractor personnel, while under escort in restricted areas, shall always obey the direction and orders of their escorts. During any time of an emergency, the Contractor personnel may be directed to vacate the area(s) and return later without additional costs to be incurred by the Government.

2.3. Restoration. The Contractor shall restore all disturbed property to the “as found” condition or better after installation. The Contractor shall comply with Base Grounds Restoration requirements.

2.4. Project Residue. The Contractor shall dispose all residues from this project off base and IAW Federal, local, and base environmental laws, and regulations. See 2.2.2 above.

2.5. Excavation/Building Penetrations: All wall penetrations, shall be restored to meet the required base fire ratings.

2.6. Labeling: The Contractor shall label all equipment and cables they install in accordance with ANSI/TIA/EIA-606-A-2002 and as directed by the POC.

2.7. Redline Drawings: The Government shall provide additional (one copy each) building floor plans and scaled JB MDL Base Maps to the Contractor after contract award. The Contractor shall provide updated drawings at project completion (CDRL A001).

2.8. Outages: The Contractor shall be responsible for preventing any unscheduled (i.e. cutting or disabling any in-service cables or equipment). Contractor-caused, interruptions of communications capabilities shall be immediately identified to the POC and PPOC. Such outages shall be Contractor worked non-stop until corrected, entirely at the Contractor’s expense. The Contractor shall coordinate planned outages with the POC at least 10 calendar days in advance of the outage if the implementation necessitates disruption of service, (e.g., communications, electrical, or other utilities).

2.9. Hours of Operation: The Contractor shall perform the services required under this contract during normal duty hours 0700-1600 of the facility where work is being performed. Any other work schedule outside of normal duty hours requires PPOC and POC approval. Part or all the facility may be occupied during this effort. Contractor shall accomplish the installation with minimum disruption of daily office activities. If performance of work outside normal duty hours will be required to meet the schedule, include desired working hours in the Contractor’s Proposal.

2.10. Holidays/Down Days: The Contractor shall not perform work under this contract on US Federal holidays or site-unique down-days unless expressly authorized by the PPOC and POC.

2.11. Final Acceptance: The Contractor shall schedule a final project walk-through with the PPOC and POC. This should be scheduled at least 5 calendar days prior to acceptance. Final acceptance will be when the Point of Contact signs off on the AF Form 747, thereby indicating Government Acceptance.

3.0. COMMODITY REQUIREMENTS: The Contractor shall EFI&T the SRS as defined herein. The 87 CS will provide all escorts for Contractor personnel to accomplish installation and testing. The Contractor shall only offer the latest version of the SRS offered by

its manufacturer. The Contractor shall **NOT** offer an SRS, or any portion thereof, that is on the Manufacturer's End of Life or End of Sales lists.

3.1. B 1901, RDP Room: This is the main Communications Demarcation for the JB MDL – refer to Table 1 (all circuits route thru/back to B 1901). The Contractor shall EFI&T the required SRS within Bay FP 1013. All FOPPS and drop circuit cross connects are within 40' cable running distance.

3.1.1. The Contractor shall EFI&T, in FP 1013, the required CTP switch (or like functional device) for analog circuit connectivity to Hangar 1. The CTP's network connection shall be terminated to the SRS here and routed, via software configuration, to the matching Hangar 1 CTP. The analog circuit connections shall be Contractor EFI&T CAT 5e.

3.1.2. The Contractor shall EFI&T a 2 KVA Uninterruptable Power Supply (UPS) in the bottom of FP 1013. There will be 110 VAC GFE female Outlets available for the Contractor to plug their UPS into.

The Contractor shall also EFI&T a 6 each 110 VAC Outlets Power Distribution Unit (PDU) Strip 1U above their UPS and EFI&T its service plug to their EFI&T UPS. The Contractor shall EFI&T all SRS and CTP power connections to this UPS.

3.2. B 2217, Switch Room: The Contractor shall EFI&T the required SRS within Bay 1004. All

FOPPs and drop circuit cross connects are within 20' cable running distance. The Contractor shall

EFI&T a 2 KVA Uninterruptable Power Supply (UPS) in the bottom of FP 1004. There will be 110 VAC GFE female Outlets available for the Contractor to plug their EFI&T UPS into. The Contractor shall also EFI&T a 6 each 110 VAC Outlets Power Distribution Unit (PDU) Strip 1U above their UPS and EFI&T its service plug to their EFI&T UPS. The Contractor shall EFI&T all SRS power connections to this UPS.

3.3. B 1730, ITN Room: The Contractor shall EFI&T the required SRS within Bay 1004. All FOPPs and drop circuit cross connects are within 20' cable running distance. The Contractor shall EFI&T a 2 KVA Uninterruptable Power Supply (UPS) in the bottom of FP 1004. There will be 110 VAC GFE female Outlets available for the Contractor to plug their EFI&T UPS into. The Contractor shall also EFI&T a 6 each 110 VAC Outlets Power Distribution Unit (PDU) Strip 1U above their UPS and EFI&T its service plug to their EFI&T UPS. The Contractor shall EFI&T all SRS power connections to this UPS.

3.4. B 5632, ITN Room: The Contractor shall EFI&T a new 7' X 19" Relay Rack (RR) for this location. The Contractor shall EFI&T all required SRS within this new RR. The exact RR locations shall be agreed upon between the Contractor Installation personnel and the POC. Once Contractor mounted to the floor, the Contractor shall ground the RR to the existing South wall mounted Ground Bar approximate distance shall not exceed 30'. The Government shall extend 110 VAC Power to the RR once it is Contractor installed/grounded. The Government shall terminate the 110 VAC feed to a Quad outlet on the lower back of RR. All FOPPs and drop

circuit cross connects are within 20' cable running distance. The Contractor shall EFI&T a 2 KVA Uninterruptable Power Supply (UPS) in the bottom this new Relay Rack. There will be 110 VAC GFE female Outlets available for the Contractor to plug their EFI&T UPS into. The Contractor shall also EFI&T a 6 each 110 VAC Outlets Power Distribution Unit (PDU) Strip 1U above their UPS and EFI&T its service plug to their EFI&T UPS. The Contractor shall EFI&T all SRS power connections to this UPS.

3.5. Hangar 1, Switch Room: The Contractor shall EFI&T the required SRS within Bay 4. The Contractor shall EFI&T the CTP in this Bay also. Again, physically and logically connect the CTP back to its mate in B 1901 via the SRS. All FOPPs and drop circuit cross connects are within 80' cable running distance. There is existing 110 VAC Technical Power in back of this Bay. The Bay is Grounded.

3.6. B 5656, Room 1311: The Contractor shall EFI&T the required SRS within Bay 002. All FOPPs and drop circuit cross connects are within 40' cable running distance. There is existing 110 VAC Technical Power in back of this Bay. The Bay is Grounded.

3.7. B 5321, DCO: The Contractor shall EFI&T the required SRS within the Bay Closest to entrance. The Bay is not named so the COR will identify it at the first part of the Contractor's EFI&T efforts. All FOPPs and drop circuit cross connects are within 80' cable running distance. There is existing 110 VAC Technical Power in back of this Bay. The Bay is Grounded.

3.8. B 3369, Switch Room: The Contractor shall EFI&T the required SRS within FP 1001. All FOPPs and drop circuit cross connects are within 40' cable running distance. There is existing 110 VAC Technical Power in back of this Bay. The Bay is Grounded.

3.9. The Contractor shall EFI&T each location's SRS unit with Main and Standby Fiber Optic Transmit and Receive modules. The Configuration shall be implemented in a dual rotating redundant Fiber Ring.

Again, all Outside Plant fiber is GFE. The Contractor shall first test their SRS Ring end-to-end before Cross connecting any Table 1 Drop Circuits. This testing shall include testing each SRS' failover capability between the Main and Standby Fiber Optic Transmit and Receive modules. The Contractor shall provide test results to the POC proving proper end-to-end connectivity. Once this testing is validated complete by the POC the Contractor shall work with 87 CS personnel and cutover the Table 1 circuits. The 87 CS personnel shall assist and participate in identifying all Table 1 circuits and their appropriate cross connections.

3.10. The Contractor shall EFI&T all material, hardware, software, and configurations as specified in the SOW and noted below. The Contractor shall coordinate all installations through the COR.

3.11. Contractor Furnished (CF) Materials and Network Specific Warranty: The Contractor shall furnish all materials for this project except as identified. The Contractor shall supply Commercial-Off-the-Shelf (COTS) end-item system components required for contract performance wherever possible. The Contractor shall identify all materials or equipment required for the project in their Proposal. The Contractor shall list the manufacturer and part

number of the proposed material in the Proposal. Unless otherwise specified, the Contractor shall provide copies of existing COTS technical or equipment manuals and warranty documentation and procedures for Contractor-furnished end-item system equipment maintenance. The Contractor shall identify and conduct system familiarization for Government personnel to ensure proper operation and maintenance of the system(s). See paragraph 5.5.

3.12. Physical Inspection: The physical inspection consists of the Contractor performing a physical inspection of each location where installation is required. At each location, the Contractor shall verify the specified placement of the equipment to ensure the installation shall meet required safety standards. The Contractor shall verify that each location shall have proper rack space, proper mounting of equipment, rack grounding, and proper primary power. An installation test report IAW CDRL A005 shall be provided after physical inspection is completed. Results shall comply before component validation and operational tests are run.

3.13. Component Validation: The Contractor shall perform component validation by testing all the individual Contractor EFI&T SRS, cables, and other related equipment to ensure they are working correctly. The Contractor shall test components prior to assembly and installation before continuing to Operational Test. An installation test report IAW CDRL A005 shall be provided after component validation is completed.

3.14. Operations Test: The operations test shall consist of testing the newly installed SRS. This shall be the real-world test to ensure functionality of the new installation and equipment related to this project. An installation test report IAW CDRL A005 shall be provided after operational test is completed.

3.15. Grounding and Bonding: The EFI&T equipment shall be grounded and bonded to the building ground. All grounding and bonding shall follow J-STD-607-A.

4.0. MANAGEMENT:

4.1. Schedule. The Contractor shall provide a complete milestone schedule that denotes major activities to include time-phased start and completion dates for this project and sub-projects associated with the installation of the components and systems. (CDRL A002).

4.2. In-Brief. The Contractor shall conduct an in-brief with the POC or designated representative and other site personnel before the start of implementation. The in-brief shall familiarize the customer with the purpose of the Contractor's visit and coordinate the project schedule (e.g., arrange for required escorts; QAP schedule; etc.).

4.3. Project Manager or Site Point of Contact (POC) on Site. The Contractor shall designate the Contractor's on-site team leader and alternate(s) as the Site POC for this project in their Access Request Letter. They shall oversee all facets of the project. The Site POC or alternate(s) shall be on site during duty hours until project completion. The Site POC shall be the interface

for all work site communications with the Government, including quality, safety, and discrepancy matters.

4.4. Quality Assurance (QA). The Contractor shall provide quality assurance support for the entire life of the project. This support shall assist the Government representative in performing random spot checks and in performing the system acceptance tests. Contractor QA shall be responsible for identifying deficiencies and/or discrepancies throughout the life of the project to the POC. A weekly report (soft copy) will be submitted indicating project progress/statuses and listings any deficiencies/discrepancies found and actions to correct them. (CDRL A003).

4.5. Compliance. The Contractor shall remain in compliance with all Federal, State, and base security and safety laws, regulations, policies, and requirements.

4.6. Clean up. At day's end, the installer shall remove all debris and surplus materials from the workplace. Equipment and materials required to complete the work effort may remain on site if they are organized/stored in a manner that does not cause a safety hazard.

4.7. Service Outages. The Contractor shall coordinate planned outages with the POC at least ten workdays in advance of the outage if the implementation necessitates disruption of service, (e.g., communications, electrical, or other utilities).

4.8. Environmental Management. The Contractor shall comply with the most stringent environmental Federal, and local laws and regulations. and Air Force policies, instructions, and plans. The Federal Government is not exempt from compliance with environmental regulations. The Contractor shall maintain an awareness of changing environmental regulatory requirements to avoid environmental deficiencies for activities on JB MDL. The Prime Contractor shall ensure their sub-Contractors (if any) comply with these specifications.

4.9. Walk-Through. The Contractor shall schedule a final project walk-through of all work completed prior to close out, with the POC. This should be scheduled at least 72 hours prior to the event.

4.10. Personnel Requirements: The Project Manager, Site POC, and respective alternate(s) shall be able to read, write, speak, and understand English. All reporting and documentation shall be in English. The Contractor's technical team shall be Vendor Certified to EFI&T the SRS. In their proposal, the Contractor shall confirm their personnel have the required certifications.

4.11. The Contractor shall coordinate with the site POC to obtain base access. The Contractor shall process and provide a site Access Request letter, to the POC, within 5 calendar days after contract award. This Access Request letter shall identify the names (as shown on the individual's Passport or driver's license), Passport or driver's license numbers, the state of issue, birth date, and citizenship of the personnel who will be performing work under this SOW. It shall also include the company name, address, phone number, contract number, project start date and end date. If required by the POC, the Contractor shall provide identification badges for their

employees. All Contractor personnel shall wear these badges while on duty on the Government site. The badges shall identify the individual, company name, and be clearly and distinctly marked as Contractor and be IAW base regulations.

5. DELIVERABLES.

5.1. List of Deliverables. All deliverables are subject to Government acceptance and approval. They shall meet professional standards and the requirements set forth in this SOW. All deliverables shall be produced using recommended software tools/versions as accepted by the Government. The Contractor shall submit the following deliverables:

CDRL	Data Item Title
A001	As Built
A002	Installation Plan
A003	Training Materials
A004	Reserved
A005	Test Report
A006	Test Plan

5.2. Cutover Plan: The Contractor shall provide a detailed cutover plan IAW CDRL A006 to delineate how the transition shall be implemented. This plan should outline how the system shall be placed into service and clearly specify how the newly installed systems shall connect to the existing base systems IAW CDRL A006. All closeout documents, including topologies, will be provided prior to project closeout.

5.3. Acceptance Test Plan: The Contractor shall provide a detailed acceptance test plan of the installation covering all aspects of the installation and demonstrate the acceptability of the work on meeting the Government's requirements. The Contractor shall perform all tests and document results of tests with Government's witnesses overseeing the Acceptance testing IAW CDRLs A005 and A006.

5.4. As-Built Drawings: The Contractor shall provide as-built drawings for the installed equipment to include where and how they connect/interface with each other and existing base systems IAW CDRL A001.

5.5. Training: The Contractor shall provide operational hands-on training for 5 Government Personnel. The Contractor shall provide all training materials (slides, Technical Documents, Manuals, etc.) IAW CDRL A003. The training shall cover complete capabilities of the SRS.

The training shall also provide Operations and Maintenance training for the complete SRS so Government personnel shall be capable of adequately maintaining and providing future additional interfaces. This training shall cover the SRS' Hardware/Software and Operations and

Maintenance functions and their uses. The POC shall arrange a training space for the trainees and Contractor trainer.

5.6. Testing Requirements: The Contractor shall perform testing and inspections of all systems solutions to ensure the technical adequacy and accuracy of all work, including reports and other documents required in support of that work. The Contractor shall conduct on-site testing in accordance with the SRS Manufacturer's installation manuals, practices, and the appropriate test procedures. The Contractor shall include testing the redundancy and failover functionality of the Contractor EFI&T SRS. The Contractor shall provide a detailed test plan IAW CDRL A006.

5.6.1. When any system, subsystem, component, or requirement test fails to meet the requirements of the test, Government acceptance shall be withheld until such time as the cause of the failure is corrected to the Government's satisfaction.

5.6.2. After appropriate corrective action has been taken, all tests including those previously completed related to the failed test and the corrective action shall be repeated and successfully completed prior to Government acceptance. Pre-cutover audits shall consist of verification of all testing completed by the Contractor such that the system is deemed ready for functional cutover. As part of this audit, any engineered changes, or approved waivers applicable to the installation shall be reviewed and agreed upon between the Contractor and the POC. Post-cutover audits shall verify that all post-cutover acceptance testing has been performed satisfactorily in accordance with the standard practices and identify those tests, if any which have not been successfully completed and shall be re-tested prior to acceptance. The Contractor shall do all testing in accordance with the acceptance test plan mentioned above. The Contractor shall provide a logical test process that minimizes interruptions and avoids sustained downtime and presents a contingency procedure to be implemented in the event of systems failure during testing.

5.6.3. The Contractor shall furnish all test equipment and personnel required to conduct all required testing. The Contractor shall document all test results in an installation test report IAW CDRL A005. During the testing phase, the Government reserves the right to perform any of the Contractor performed inspections and tests to assure solutions conform to prescribed requirements. The Contractor shall provide on-site support during the acceptance testing. The Test Report(s) shall be provided to the Government no later than 10 days after test(s) have been completed.

5.7. System Acceptance: Acceptance of the SRS shall be dependent upon successful completion and documentation of tests in the Contractor submitted/Government approved test plan IAW CDRL A006 and test report CDRL A005. Screen shots (if applicable) shall serve to document successful test results.

6.0. WARRANTY. The Contractor shall provide a one-year warranty or manufacturer's standard commercial warranty, whichever is longer. The Contractor shall also provide a one-year workmanship warranty. The warranty period shall start from the date of system and or project acceptance (signing of AF Form 747 by the Point of Contact). The Contractor shall provide hardware replacement, major and minor software upgrades for one year for all TC Comm and UPS equipment. The Contractor shall transfer warranties to the Government.

7.0. PERFORMANCE.

7.1. Period of Performance. The Contractor shall have 1 year from contract award to complete all facets of the project including final walk through with the Government representative. The period of performance for this contract shall consist of one (1) base year.

7.2. Place of Performance. The place of performance is JB MDL NJ.

7.3. Hours of Work. The Contractor shall routinely work during normal operating hours of the site. Monday through Friday (0700-1600 hrs.), excluding Federal Holidays. Any site work requested by the Contractor to be performed outside of normal duty hours shall be coordinated with the POC at least 10 working days in advance.

8.0. GOVERNMENT FURNISHED INFORMATION AND STANDARDS.

The Contractor shall comply with all base installation standards. The Contractor shall comply with all commercial codes and standards where applicable in the performance of this contract. It is the Contractor's responsibility to identify and obtain applicable standards and best commercial practice documentation. Should a conflict arise between this SOW and any commercial standard referenced, the Contractor shall make the Government aware of the conflict. The Government shall determine which standard applies.

8.1. Reference Standards:

NUMBER	TITLE	WEBSITE OR LOCATION
FIPS	Federal Information Processing Standards Publications	https://csrc.nist.gov/publications/fips
BICSI	Building Industry Consulting Service International, Inc. (BICSI)	http://www.bicsi.org/

NFPA 70 (2020)	National Fire Protection Association (NFPA), National Electric Code (NEC)	http://www.nfpa.org/
ANSI	American National Standards Institute (ANSI) and Internet Standards Resources	http://www.ansi.org/
TIA	Engineering: TIA Standards See Commodity Specific Appendices and Individual Task Orders	http://www.tiaonline.org/
ANSI/TIA-569-E (2019)	Commercial Building Standards for Telecommunications Pathways and Spaces	http://www.tiaonline.org/
ANSI/TIA-570 (2008)	Residential Telecommunications Cabling Standard	http://www.tiaonline.org/
EIA/CIA-310-E (2005)	Racks, Panels and Associated Equipment	http://global.ihs.com/
ANSI/NEMA-FB 1 (2012)	Fittings, Cast Metal Boxes and Conduit Bodies for Conduit, Electrical Metallic Tubing (EMT) and Cable	http://www.nema.org
ANSI/EIA-310-D	Cabinets, Racks, Panels, and Associated Equipment	http://www.server-racks.com/eia-310.html
ANSI/TIA/EIA-606-C – (2017)	Administration Standard for the Telecommunications Infrastructure of Commercial Buildings	www.flexcomm.com/library/606aguide.pdf
ANSI-J-STD-607 – D (2019)	Commercial Building Grounding (Earthing) and Bonding Requirements for Telecommunications	http://www.ansi.org/
BICSI TDMM Manual - (2006, 11 th Edition)	Building Industries Consulting Services International Telecommunications Distribution Methods (TDM) Manual	https://www.bicsi.org/default.aspx
NEMA TC 2- (2013)	Electrical Polyvinyl Chloride (PVC) Tubing and Conduit	http://www.nema.org/Standards/ComplementaryDocuments/Contents%20and%20Scope%20NEMA%20TC%202-2013.pdf

OSHA CFR 29 Part 1910-268 -	Telecommunications	https://www.osha.gov/pls/oshaweb/owasrc.h.search_form?p_doc_type=STANDARDS&p_toc_level=1&p_keyvalue=1910
TIA 526-14-B (2010)	Optical Power Loss Measurement of Installed Multimode Fiber Cable Plant	http://www.tiaonline.org/
REA TE&CM 643 Form 515C	Rural Electrification Administration (REA)	http://www.rurdev.usda.gov/supportdocuments/utp_bulletins_1753f-152.pdf
ANSI/TIA-568-.1. E (2020)	Commercial Building Telecommunications Cabling Standard	http://www.tiaonline.org/
ANSI/TIA-568- .2.D (2018)	Balanced Twisted-Pair Telecommunication Cabling and Components Standard	http://www.tiaonline.org/
ANSI/TIA-568- .3.D (2016)	Optical Fiber Cabling Components Standard	http://www.tiaonline.org/
ANSI/TIA-570 (2008)	Residential Telecommunications Cabling Standard	http://www.tiaonline.org/
ANSI/TIA-598-C (2005)	Optical Fiber Cable Color Coding	http://www.tiaonline.org/

9.0. SAFETY.

The Contractor shall be responsible for complying with all Federal, State, and Base safety guidelines.

9.1. Accident/Incident Reporting and Investigation: The Contractor shall record and report all available facts relating to each instance of accidental damage to Government property or injury to either Contractor or Government personnel to the Base Safety Office and POC. The Contractor shall secure the scene of any accident and wreckage until released by the accident investigative authority through the Contracting Officer. If the Government elects to investigate the incident, the Contractor shall cooperate fully and assist the Government personnel until the investigation is completed.

9.2. Potential Problems: The Contractor shall determine any potential problems with the projected work in areas that may contain asbestos, hazardous materials, and hazardous wastes and local procedures to be followed in the event such problems are encountered.

9.3. Disposal/Reporting: The Contractor shall comply with Federal, State, and Base procedures concerning the use, disposal, or reporting of hazardous materials.

9.4. Hazardous Material Identification: The Contractor shall submit Material Safety Data Sheets (MSDS) for any hazardous materials (paints, solvents, cleaners, encapsulating compound, etc.) to be used in performance of the contract. The Contractor shall file the MSDS(s) with the Base Environmental Management Office prior to using hazardous material. The POC may assist with this effort.

9.5. Identification / Marking: The Contractor shall clearly mark all Contractor Furnished Packaging (CFP) with their company's name, contract number, and project number. The Contractor shall place an easily read, very visible, sign (minimum 8.5" x 11") on large containers, construction equipment, or un-manned rental vehicles while on the Government installation indicating the company name; contract/task order number; project title; and both the Contractor and Site POC's names and local telephone numbers.

9.6. Construction Equipment: The Contractor shall provide construction equipment (e.g., man lifts, etc.) required to support implementation. The Contractor shall be responsible for transportation of project materials and construction equipment to/from the work site, as well as their security.

9.7. Transportation and Storage of Material/Equipment: The Contractor shall transport and store all Contractor Furnished Materials (CFM) and Contractor Furnished Equipment (CFE) required for completion of the contract requirements.

10. List of Attachments: N/A